Emir Ulurak

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M.Sc. Student in Artificial Intelligence, Özyeğin University

Research interests: 3D computer vision – reconstruction, scene understanding and world models.

Education

Özyeğin University — **Istanbul, Turkey**
M.Sc. in Artificial Intelligence — *09/2024 – 06/2027 (expected)*

- Cumulative GPA: 3.96 / 4.00
- Thesis: *MonoMIP: Multi-Height Inverse Perspective Mapping for Monocular 3D Object Detection*
  Supervisor: Mustafa Furkan Kıraç

Özyeğin University — **Istanbul, Turkey**
B.Sc. in Computer Science — *09/2019 – 06/2024*

- Cumulative GPA: 3.24 / 4.00 – graduated with Honors
- Senior project: *Diffusion-Based Valuable NFT Generation*
  Supervisor: Emre Sefer

Publications

MonoMIP: Multi-Height Inverse Perspective Mapping for Monocular 3D Object Detection
Emir Ulurak, Mustafa Furkan Kıraç
Asian Conference on Computer Vision (ACCV), 2026. To appear. [paper] [code]

Diffusion-Based Valuable NFT Generation
Emir Ulurak[*], Beyza Kaya[*], Emre Sefer
Multimedia Tools and Applications, 85:557, 2026. [doi] [pdf] [code]

Copilot-Driven Development in Practice: Building a Monolith and Migrating to Microservices
Arif Kaan Bitgin, Orcun Demir, **Emir Ulurak**, Mustafa Furkan Kıraç, Hasan Sözer
European Conference on Software Architecture (ECSA), Industrial Track, 2026. [page]

[*]Equal contribution.

Experience

Özyeğin University, VGL – Computer Vision and Graphics Lab — **Istanbul, Turkey**
Graduate Researcher – Supervisor: Mustafa Furkan Kıraç — *09/2024 – present*

MonoMIP: Multi-Height Inverse Perspective Mapping for Monocular 3D Object Detection | ACCV 2026

- Addressed the misprojection of above-ground objects in single-plane inverse perspective mapping (IPM).
- Key insight: when image features are projected onto horizontal planes at several heights, the way an object's features change from plane to plane depends on its depth, giving a geometric cue for depth and occupancy.
- Projected features onto four planes, from the ground to roof height, and correlated each upper plane with the ground plane; a bird's-eye-view decoder reads these alongside a depth-guided image decoder.
- Achieved 21.66 $AP_{3D|R40}$ (Car, Moderate) on the KITTI test set, the best among published monocular methods without extra training data (as of 10/2026); released code and checkpoints.
- Designed and implemented the method, ran all experiments and wrote the paper.

Joint Uncertainty for Feed-Forward 3D Reconstruction | Ongoing, 2026

- Showed that the errors of five feed-forward 3D models (VGGT, Pi3, MapAnything, MASt3R, DUSt3R) are strongly correlated between nearby points on ScanNet++, ARKitScenes and Waymo; treating pointwise errors as independent therefore produces miscalibrated uncertainty for measurements that combine several points, underestimating whole-scene errors and overestimating errors on short distances.
- Built a joint uncertainty model on a frozen reconstruction model that samples the errors of all queried points together; on 50 ScanNet++ test scenes it reduced the Brier score for four-point groups by 25% and the interval score for angles by 22% relative to calibrated VGGT confidence.
- Evaluated with proper scoring rules (interval, Brier and energy scores) against equally trained baselines, including Trust3R adapted to VGGT, with significance tests. Ongoing: a learned generative error field and transfer to new datasets.

Özyeğin University – TÜBİTAK 1505 Project **Istanbul, Turkey**
M.Sc. Scholar – PI: Hasan Sözer *04/2026 – present*

- Project: *Transformation of Monolithic Systems into Microservices with Hypergraphs and AI Models*.
- Designed an evaluation framework that replays an application's real historical changes as AI coding-agent tasks to judge a service decomposition; observed lower agent task success after service decomposition in the evaluated tasks.
- Built a Java/Spring migration workflow combining static analysis, deterministic refactoring and local LLMs (4–27B), keeping source code local and checking changes against existing tests.

Özyeğin University **Istanbul, Turkey**
Undergraduate Researcher (Senior Project) – Supervisor: Emre Sefer *10/2023 – 06/2024*

- Fine-tuned Stable Diffusion with reinforcement learning (PPO) to generate NFT images with rare visual attributes while preserving prompt fidelity.
- Designed a rarity-aware reward from a ViT-B/16 rarity classifier, combined with ImageReward for visual quality; KL regularisation towards the supervised fine-tuned model balances rarity against visual coherence.
- Evaluated rarity and visual quality, including reward-weight ablations; equal-contribution first author.

Fishency Innovation AS **Remote (Istanbul, Turkey)**
Research and Development Intern *08/2023 – 09/2023*

- Worked on underwater imaging for salmon farming, where a multi-view stereo camera setup is used for fish biomass estimation and fish-health monitoring.
- Applied self-supervised pre-training to few-label fish segmentation; explored temporal cues for fish tracking.

INOVAKO **Istanbul, Turkey / Vitoria-Gasteiz, Spain**
Computer Vision and AI Intern *06/2023 – 07/2023*

- Developed a multi-camera machine-vision system (8 Basler industrial cameras) for automated surface-defect inspection (holes, cracks) of reflective automotive parts on a production conveyor.
- Tuned per-camera exposure and trigger timing in a dark-room setup to suppress specular reflections.
- Built the detection pipeline: data collection and annotation, YOLO fine-tuning and TensorRT-accelerated inference.

Özyeğin University **Istanbul, Turkey**
Volunteer Research Assistant *10/2022 – 06/2023*

- Ocean-eddy segmentation from flow-field data with transformer and diffusion models, jointly predicting masks and vortex direction and magnitude; with PhD researchers at Rutgers University.
- YOLO-based object detection on aerial imagery (xBD, VisDrone).

Teaching Experience

Özyeğin University **Istanbul, Turkey**
Teaching Assistant *09/2024 – present*

- **CS104 – Python Programming (about 200 students):** Main teaching assistant; led lab sessions teaching students to code by solving problems live, managed the quizzes and graded the midterm examinations.
- **CS321 – Programming Paradigms:** Graded midterms; building a proctored coding platform with local AI agents.
- **CS304 – Introduction to AI, Machine Learning and Data Science:** Designed and graded course projects.
- **CS452/552 – Data Science with Python:** Graded course projects.
- **CS409/509 – Advanced C++ Programming:** Graded examinations.

Skills and Languages

Programming	Python, C++
Tools	PyTorch, Slurm, Blender, Docker, LaTeX
Languages	Turkish (native), English (professional working proficiency)